

BRFSS Mental Health Prediction Using Machine Learning

Data and Study Population

This study used data from the **Behavioral Risk Factor Surveillance System (BRFSS)** collected over seven years, from **2018 to 2024**. The preprocessed dataset consisted of **19,999 observations**, with **20 predictor variables** and one target variable.

The target variable was **MENTHLTH**, which represents the number of days during the past 30 days on which a respondent reported that their mental health was not good. The variable ranges from 0 to 30 days and was modeled as a continuous outcome.

The 20 predictor variables included demographic, socioeconomic, behavioral, and health-related characteristics. The variables used in the analysis are summarized below.

Variable	Description
DECIDE	Difficulty concentrating, remembering, or making decisions
_AGE80	Imputed age value, with ages above 80 collapsed
PHYSHLTH	Number of days physical health was not good
GENHLTH	General health status
_BMI5	Calculated body mass index (BMI)
DIFFALON	Difficulty doing errands alone
MARITAL	Marital status
_STATE	State FIPS code
YEAR	Survey year
HHADULT	Number of adults in household
_IMPRACE	Imputed race/ethnicity
_SMOKER3	Four-level smoking status
FMONTH	File month

RENTHOM1	Home ownership or rental status
EMPLOY1	Employment status
_PHYS14D	Three-level physical health status based on number of unhealthy days
QSTLANG	Questionnaire language
QSTVER	Questionnaire version
_EDUCAG	Calculated education level
_ASTHMS1	Computed asthma status

The outcome variable was:

Variable	Description
MENTHLTH	Number of days mental health was not good during the past 30 days

Data Preparation

Before model training, the target variable, **MENTHLTH**, was separated from the predictor variables. Identification and survey administration variables that were not intended to serve as predictors were excluded, including **SEQNO**, **_PSU**, **_STSTR**, **IDATE**, **IMONTH**, **IDAY**, and **IYEAR**.

Variables with the potential to introduce direct target leakage were also excluded. For the prediction of **MENTHLTH**, these variables included **_MENT14D** and **POORHLTH**. In addition, survey weighting and design variables, including **_LLCPWT**, **_LLCPWT2**, **_STRWT**, **_WT2RAKE**, and **_RAWRAKE**, were excluded when present.

These preprocessing steps were applied to reduce the possibility that model performance would be influenced by identification variables, survey weights, or variables containing direct information about the target.

Training and Test Data

The dataset was randomly divided into training and testing subsets using an **80/20 train-test split**. Of the 19,999 observations, **15,999 observations were used for model training**, while the remaining **4,000 observations were reserved as an independent test set**.

A random seed of **42** was used to ensure reproducibility of the train-test split.

For regression models that are sensitive to the scale of predictor variables, including Ridge, Lasso, and Poisson regression, the predictors were standardized using **StandardScaler**. The scaler was fitted using only the training data and subsequently applied to the test data.

Machine-Learning Models

Six regression algorithms were evaluated for predicting the number of mentally unhealthy days:

1. XGBoost regression
2. Random Forest regression
3. LightGBM regression
4. Ridge regression
5. Lasso regression
6. Poisson regression

The use of multiple algorithms allowed comparison between nonlinear tree-based ensemble approaches and more traditional regression-based methods.

Hyperparameters were optimized using **5-fold cross-validation**, with **mean absolute error (MAE)** as the primary optimization criterion. The training data were divided into five subsets, with four subsets used for model training and the remaining subset used for validation. This procedure was repeated across all five folds to obtain a more robust estimate of model performance.

Model Evaluation

Model performance was evaluated using three regression metrics: **mean absolute error (MAE)**, **root mean squared error (RMSE)**, and **coefficient of determination (R^2)**.

MAE represents the average absolute difference between the observed and predicted number of mentally unhealthy days. Lower MAE values indicate better predictive performance.

RMSE also measures the difference between observed and predicted values but places greater weight on larger prediction errors. Therefore, lower RMSE values indicate better model performance.

R^2 represents the proportion of variability in **MENTHLTH** explained by the model. Higher R^2 values indicate greater explanatory performance.

All final performance metrics were calculated using the independent test dataset.

Model Performance

The performance of the six machine-learning models is summarized below.

Model	CV MAE	Test MAE	Test RMSE	Test R ²
XGBoost	4.3157	4.4033	7.0942	0.2856
Random Forest	4.3906	4.4747	7.1672	0.2708
LightGBM	4.4242	4.4829	7.1494	0.2744
Ridge Regression	4.4790	4.5600	7.1428	0.2757
Lasso Regression	4.4783	4.5582	7.1426	0.2758
Poisson Regression	4.4227	4.5453	7.3250	0.2383

Among the evaluated models, **XGBoost achieved the lowest test MAE and was therefore selected as the best-performing model**. XGBoost achieved a cross-validation MAE of **4.3157 days**, a test MAE of **4.4033 days**, an RMSE of **7.0942 days**, and an R² of **0.2856**.

The test MAE indicates that the XGBoost predictions differed from the observed number of mentally unhealthy days by approximately **4.4 days per respondent, on average**. The R² value of 0.2856 indicates that the model explained approximately **28.6% of the variability in MENTHLTH**.

The relatively small difference between the cross-validation MAE (4.3157) and test MAE (4.4033) suggests that model performance was reasonably consistent between the cross-validation and independent test datasets.

Overall, the tree-based models, particularly XGBoost, showed slightly better predictive performance than the traditional regression models for predicting the number of mentally unhealthy days in the BRFSS dataset.

Comparison of My Results with the Article

<https://www.nature.com/articles/s41598-024-84137-1>

Item	Your study	Oparina et al. (2025)
Outcome	MENTHLTH: mentally unhealthy days, 0–30	Life satisfaction / subjective wellbeing
U.S. dataset	BRFSS	Gallup Daily Poll
Years	2018–2024	2010–2018
Sample	19,999	>1 million across Germany, UK, and U.S.
Predictors	20 selected variables	Restricted and extended feature sets
Train/test split	80/20	80/20
Cross-validation	5-fold	4-fold
Tree models	XGBoost, RF, LightGBM	Gradient Boosting, RF
Linear models	Ridge, Lasso, Poisson	OLS, LASSO
Main metric	MAE, RMSE, R ²	Primarily out-of-sample R ²
Best model	XGBoost	Gradient Boosting
Best R ²	0.2856	Approximately 0.25–0.31 with extended predictors